

# **Lecture 5.5 — Companion Figures**

Calcium Dysregulation (Part 1)

Visual references for the Notion revision page. Figures reproduced from the original lecture slides (Prof Niamh Martin, 13 October 2025).

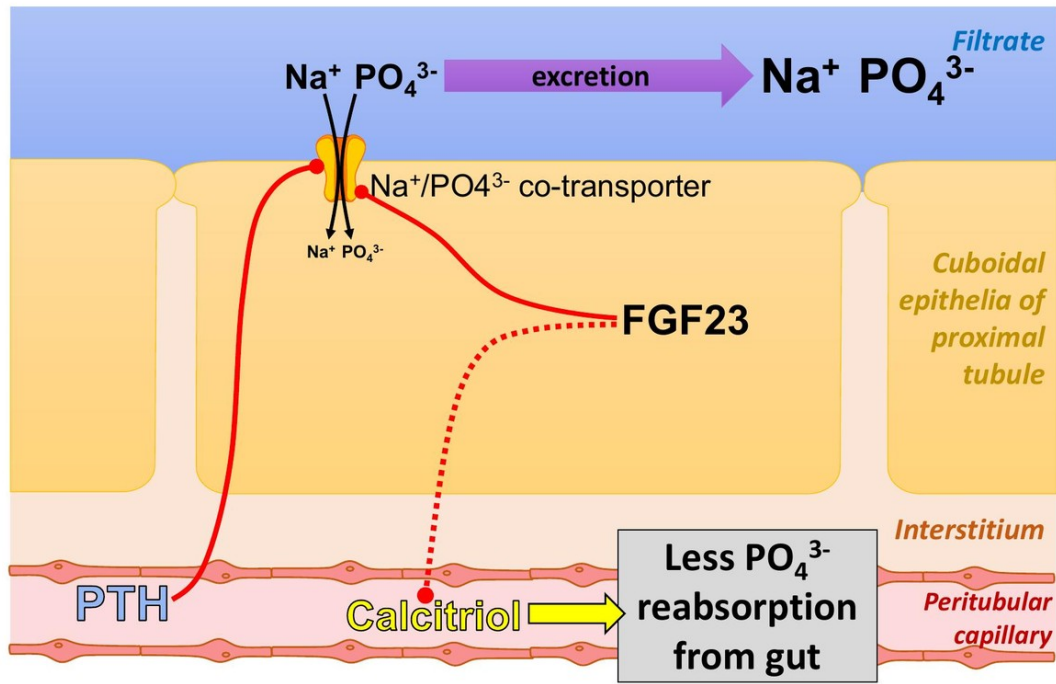


Figure L5.5.1 — FGF23 and PTH inhibit the  $\text{Na}^+/\text{PO}_4^{3-}$  cotransporter at the proximal tubule, driving phosphate excretion

## Chvosteks' sign – facial paresthesia



## Trousseau's sign – carpopedal spasm



A



B

Figure L5.5.2 — Clinical signs of hypocalcaemia: Chvostek's sign (facial nerve tap) and Trousseau's sign (carpopedal spasm after BP cuff inflation)

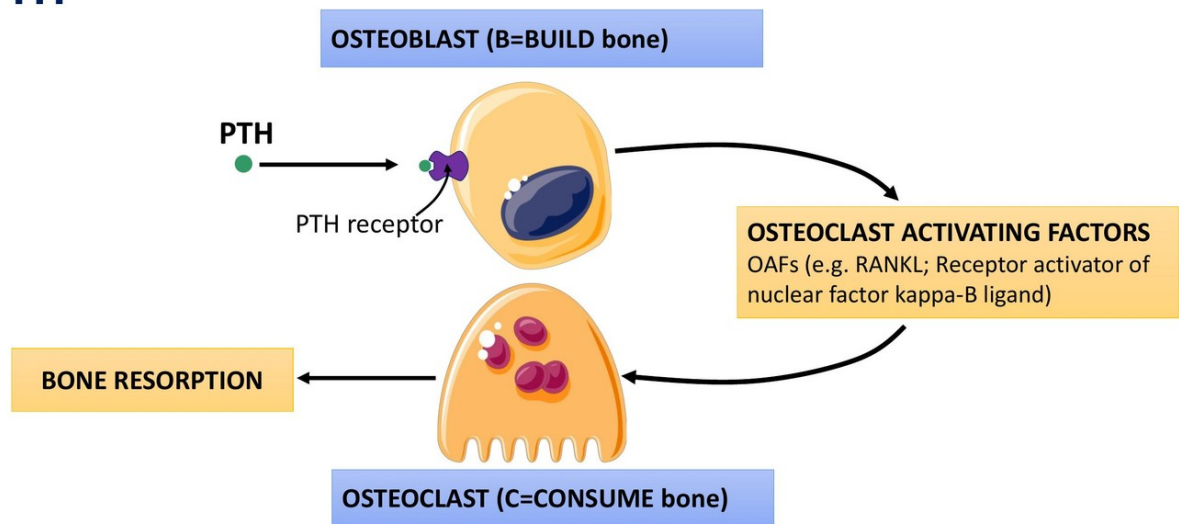
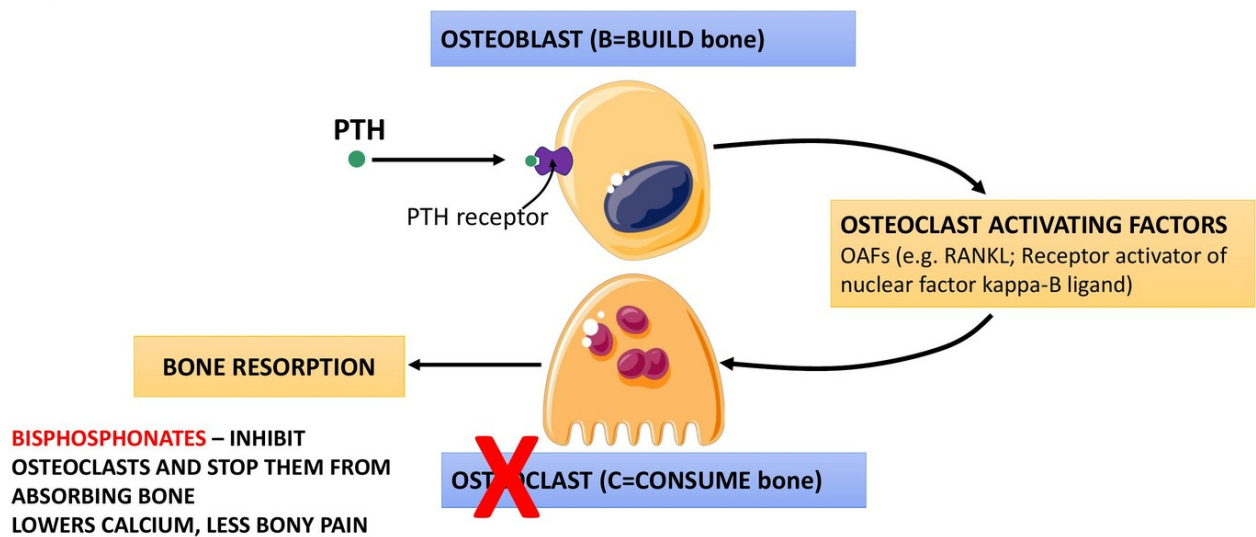


Figure L5.5.3 — PTH signals via osteoblasts (which express the PTH receptor) to release RANKL, activating osteoclasts and driving bone resorption



Endocrine disorders: Summarize the pathophysiology and pathophysiology of endocrine disorders

Figure L5.5.4 — Bisphosphonates inhibit osteoclasts, blocking bone resorption — management of hypercalcaemia of malignancy